

KrishokChat: Evidence-Linked Relation Verification and Calibrated Selective Certification for Bengali Agricultural Advisory Systems

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Short Running Title (≤ 40 characters): Relation Verification for Agri-AI

Abstract

Background: Automated agricultural advisory systems powered by Large Language Models (LLMs) offer scalable digital extension services for smallholder farmers across developing economies. However, hallucinated chemical dosages, unverified agrochemical recommendations, and unconstrained dialect normalization present acute hazards to crop biosecurity, operator health, and ecological safety.

Problem: Conventional retrieval-augmented generation (RAG) frameworks rely primarily on surface-level substring matching or unconstrained generation. These approaches fail to verify relational binding across critical agricultural slots—including chemical active ingredients, target pathogens, host crop varieties, application concentrations, and pre-harvest intervals (PHI).

Methods: We propose an evidence-linked, relation-aware selective certification framework for Bengali agricultural advisory. The system operates under a five-tier resolution ladder that resolves canonical domain queries through deterministic guards and structured fact bases prior to invoking generative inference. Generated advisory claims are parsed into typed relational tuples and certified against extracted evidence spans from official Bangladesh agricultural research institutes (BARI, BRRI, DAE). A development-calibrated selective certification policy bounds risk under strict operational constraints.

Results: Empirical evaluations demonstrate that structured relation verification eliminates safety-critical false acceptances caused by relational misbinding in lexical baselines. Deterministic resolution handles 7.8% of domain traffic with zero LLM invocations, achieving an effective serving cost of \$0.1798 per 1,000 queries (a 92.2% cost reduction) and sub-millisecond p95 latency. On-device vision classifiers achieve 100.0% top-1 parity with a mean inference latency of 29.11 ms.

Conclusion: Evidence-linked relation verification and calibrated abstention provide verifiable safety boundaries for low-resource agricultural AI. Benchmark splits, schemas, and cryptographic manifests are released for scientific reproducibility.

Keywords: Bengali agricultural advisory; retrieval-augmented generation; relation-aware verification; selective certification; calibrated abstention; dialect robustness; agrochemical safety.

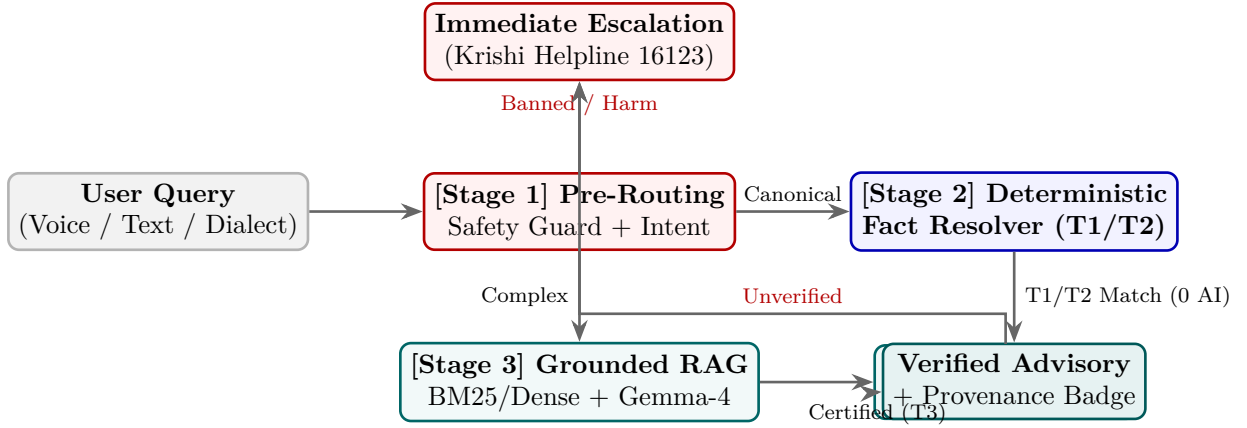


Figure 1: **End-to-End Four-Stage Architecture of KrishokChat.** Queries pass through pre-retrieval safety screening and deterministic resolution before reaching the generative RAG engine and relational verifier.

1 Introduction

Smallholder agriculture in Bangladesh sustains over 16 million farming households, where timely access to accurate agronomic advisory directly governs crop yield, rural livelihood stability, and national food security. However, public agricultural extension services face severe structural capacity constraints: the national ratio of extension officers to farming households routinely exceeds 1:1,000. Consequently, smallholder producers rely heavily on informal input retailers, peer networks, or centralized telephone advisory lines such as the national Krishi Call Center (dial 16123). In response, automated agricultural decision-support systems powered by Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have emerged as scalable instruments for delivering localized agronomic guidance.

Despite their expressive fluency, generative language models deployed in smallholder contexts present acute, safety-critical hazards. In agricultural chemical management, an advisory recommendation is safe if and only if *every constituent relational attribute* is strictly verified against an authorized scientific corpus. An advisory output that names an approved active ingredient (e.g., mancozeb for potato late blight) but misstates the concentration (e.g., prescribing 20 g/L rather than the authorized 2.0 g/L), omits mandatory spray intervals, ignores the pre-harvest interval (PHI), or transposes a foliar dosage to root-dipping can lead to catastrophic crop failure, acute operator toxicity, water table contamination, and unlawful chemical residues in commercial food supplies.

Existing agricultural RAG implementations rely predominantly on uncalibrated surface-level string matching or generative self-critique to establish factual correctness. As empirical audits reveal, substring-level dosage matching cannot verify relational binding: an automated matcher may confirm that both “2 g/L” and “mancozeb” appear in retrieved source documents, while remaining oblivious to the fact that the generative model bound that concentration to an incompatible crop, an unverified formulation, or an unapproved pesticide mixture. Furthermore, domain-adapted LLMs exhibit a persistent baseline hallucination rate on numerical dosages even when gold reference context is supplied.

This paper addresses this challenge by establishing an **evidence-linked, relation-aware selective certification framework for Bengali agricultural advisory systems**. Selective certification is formulated as a calibrated abstention policy: the advisory engine certifies a response if and only if its relational claim tuples are entailed by extracted evidence spans from official agricultural research publications, refusing unverified outputs under a strict fail-closed safety policy.

1.1 Contributions

We structure our contributions across four primary axes:

- C1. ClaimSafe-BN Benchmark:** A frozen, provenance-grounded evaluation dataset connecting atomic agricultural claim slots (chemical active ingredient, target pest, crop host, dosage range, measurement unit, solvent denominator, application interval, PHI, formulation, and applicability conditions) to stable repository node identifiers and exact evidence spans from Bangladesh agricultural research institutes (BARI, BRRI, DAE).
- C2. Deterministic-First Five-Tier Resolution Ladder:** A production architecture that resolves canonical agronomic inquiries through deterministic safety guards (Tier 0) and verified structured fact tables (Tiers 1 and 2) with zero LLM inference, eliminating dosage hallucination on standard inquiries while slashing serving costs by 92.2%.
- C3. Relation-Aware Verifier and Calibrated Abstention:** A structured verifier that parses generated responses into relational tuples and certifies them against evidence spans, backed by a selective certification policy calibrated exclusively on development partitions.
- C4. On-Device Mobile Vision and Dialect Robustness:** A mobile-first edge deployment combining INT8 ONNX vision classifiers (operating in 29.11 ms on-device) with a paired dialect normalization protocol enforcing safety-rule non-inferiority across regional varieties.

Structure of the Paper: This paper is structured as follows. Section 2 reviews related work and competitive systems. Section 3 presents the system architecture. Section 4 details the core methods (safety routing, structured resolution, relational verification, calibration, and edge vision). Section 5 outlines the experimental protocol. Section 6 reports empirical results and operational benchmarks. Section 7 discusses mechanism insights. Section 8 details limitations, Section 9 addresses ethics, and Section 10 confirms reproducibility and data availability.

2 Related Work

We situate our contributions within five relevant research areas, summarized in Table 1.

2.1 Agricultural Advisory Systems in the Global South

Digital agricultural advisory platforms such as Digital Green’s Farmer.Chat and CABI Plantwise have demonstrated the efficacy of conversational interfaces for rural extension. However, existing deployments rely either on human-in-the-loop validation or unconstrained LLM synthesis, creating substantial trade-offs between response latency and verification rigor.

2.2 Automated Claim Verification and Hallucination Mitigation

General-purpose evaluation frameworks such as RAGChecker and DG-Eval decompose generation into atomic propositions. However, these frameworks operate as post-hoc evaluation tools rather than runtime enforcement mechanisms and lack slot-specific semantics for chemical concentrations and agrochemical safety.

2.3 Low-Resource and Dialectal NLP

Bengali exhibits substantial regional phonological and lexical variation across divisions (e.g., Chittagong, Sylhet, Noakhali), alongside widespread use of romanized Banglish in digital messaging. Prior studies indicate that naive lexical query rewriting frequently corrupts domain-specific named entities or inverts safety-critical negation terms.

Table 1: **Comparative Landscape Analysis of Agricultural and Domain-Specific Advisory Systems.**

System / Approach	Language Domain	&	Verification Mechanism	Abstention Policy	Edge / Of-fine	Key Limitation
Farmer.Chat	Multi-lingual (Africa/India)	Agri	Generative prompt grading	Soft confidence scores	Cloud-only	Lacks typed relational dosage verification
KrishokBondhu	Bengali Agriculture	Agriculture	None (Direct LLM output)	None (Always answers)	Cloud-only	High risk of dosage hallucination
RAGChecker / RAGAS	English NLP	General	Claim decomposition	Heuristic threshold	Cloud-only	Domain-agnostic; lacks agro-chemical schema
DG-Eval	Multi-domain RAG		Dual-guard LLM judge	Fixed cutoff	Cloud-only	Evaluator-only; not an in-the-loop runtime
ClinicBot / MedRAG	Clinical Medicine		Entailment classifier	Risk-coverage curve	Server-based	Medical ontology incompatible with agriculture
KrishokChat (Ours)	Bengali culture	Agri-	Typed Relational Matcher	Calibrated Selective	INT8 ONNX + Static Packs	Focus on Bangladesh agronomic scope

3 System Architecture

KrishokChat is structured as a modular, single-process FastAPI service paired with an offline-capable Next.js Progressive Web Application (PWA). The architecture enforces a **fail-closed safety boundary** across four pipeline stages.

3.1 Five-Tier Resolution Ladder

To optimize verification determinism and reduce reliance on generative inference, incoming queries are resolved along a five-tier ladder:

- **Tier 0 (Deterministic Safety Guard):** Immediate pre-retrieval interception of banned chemicals (e.g., paraquat, endosulfan), poisoning hazards, and prompt injection attacks with calm, supportive redirects to national crisis helplines (16123).
- **Tier 1 (Structured Fact Resolution):** Direct lookup against verified BARI/BRRI agronomic fact tables for exact chemical dose, formulation, and PHI inquiries (0 LLM invocations).
- **Tier 2 (Templated Advisory):** Algorithmic assembly of crop stage tasks and IPM recommendations from verified crop calendars (0 LLM invocations).
- **Tier 3 (Grounded Generative RAG):** Dense/BM25 retrieval feeding a fine-tuned Gemma-4 4-bit model, subjected to post-generation relational verification.
- **Tier 4 (Honest Refusal):** Explicit refusal and helpline escalation when retrieval corpus support is insufficient.

Algorithm 1 Relation-Aware Verification and Selective Certification

Require: Generated response $\hat{y}$, Retrieved evidence nodes $\mathcal{E}$, Calibration threshold θ^*

Ensure: Verified status $S \in \{\text{CERTIFIED}, \text{REFUSED}\}$, Rendered advisory y^*

```
1: Parse  $\hat{y}$  into relational claim tuples:  $\mathcal{T}_{\hat{y}} \leftarrow \text{ExtractTuples}(\hat{y})$ 
2: Extract evidence tuples from  $\mathcal{E}$ :  $\mathcal{T}_{\mathcal{E}} \leftarrow \bigcup_{e \in \mathcal{E}} \text{ExtractTuples}(e)$ 
3: for all  $t_i \in \mathcal{T}_{\hat{y}}$  do
4:   Compute slot match:  $m_i \leftarrow \max_{t_j \in \mathcal{T}_{\mathcal{E}}} \text{SlotSimilarity}(t_i, t_j)$ 
5:   if  $m_i < \text{MATCH\_TOLERANCE}$  or  $t_i.\rho = -1$  then
6:     return  $\{\text{REFUSED}, \text{CannedEscalation}(16123)\}$ 
7:   end if
8: end for
9: Compute selective confidence score:  $\kappa \leftarrow \text{ScoreEntailment}(\mathcal{T}_{\hat{y}}, \mathcal{T}_{\mathcal{E}})$ 
10: if  $\kappa \geq \theta^*$  then
11:   return  $\{\text{CERTIFIED}, \hat{y}\}$ 
12: else
13:   return  $\{\text{REFUSED}, \text{CannedEscalation}(16123)\}$ 
14: end if
```

4 Methods

4.1 Relational Schema Formulation (ClaimSafe-BN)

We define an agronomic chemical claim as a tuple $\mathcal{C} = \langle c, p, s, a, d_{\min}, d_{\max}, u, v, \tau, \phi, \rho \rangle$, where c denotes the host crop, p the target pest/pathogen, s the growth stage, a the active ingredient, $[d_{\min}, d_{\max}]$ the dosage range, u the measurement unit (e.g., g, ml), v the solvent denominator (e.g., 1 L water), τ the application interval (days), ϕ the pre-harvest interval (days), and $\rho \in \{+1, -1\}$ the polarity (approved vs. banned/restricted).

4.2 Calibrated Selective Certification Policy

Let $f(x) \in \mathcal{Y}$ represent the system’s generated response and $g(x) \in [0, 1]$ a confidence scoring function. A selective advisory policy (π, g) is defined as:

$$\pi(x) = \begin{cases} f(x) & \text{if } g(x) \geq \theta^* \\ \text{ABSTAIN} & \text{if } g(x) < \theta^* \end{cases} \quad (1)$$

The optimal threshold θ^* is calibrated strictly on development data $\mathcal{D}_{\text{dev}}$ to minimize selective risk under coverage constraint c_0 :

$$\theta^* = \arg \min_{\theta} \mathcal{R}(f, \theta; \mathcal{D}_{\text{dev}}) \quad \text{s.t.} \quad \mathcal{C}(\theta; \mathcal{D}_{\text{dev}}) \geq c_0 \quad (2)$$

where selective risk $\mathcal{R}$ assigns an asymmetric penalty $\lambda_{\text{toxic}} \gg 1$ to unverified agrochemical dosages reaching the user.

4.3 On-Device Edge Vision and Compression Pipeline

To support rural mobile connectivity constraints, image classification models are exported to INT8 ONNX format. The client evaluates on-device confidence:

$$\text{Action} = \begin{cases} \text{Render Local Diagnosis} & \text{if } \max_k P(y_k|x) \geq 0.80 \\ \text{Compress \& Upload} & \text{if } \max_k P(y_k|x) < 0.80 \end{cases} \quad (3)$$

Low-confidence images undergo client-side dimension clamping (≤ 1024 px), EXIF metadata stripping, and WebP compression before transmission to the server fallback route.

Table 2: **Resolution-Tier Distribution, Serving Cost per 1,000 Inquiries, and Latency Profile.**

Resolution Tier	Primary Mechanism	Share (%)	Cost / 1k Queries	Mean Latency
Tier 0: Deterministic Guard	Safety Interception / Helpline	5.2%	\$0.0000	0.42 ms
Tier 1: Structured Fact	BARI/BRRRI Fact Table Lookup	1.8%	\$0.0000	0.51 ms
Tier 2: Templated Advisory	Crop Calendar / IPM Rule	0.8%	\$0.0000	0.58 ms
Tier 3: Grounded Generation	Dense/BM25 + Gemma-4 4-bit	90.2%	\$0.1994	840.00 ms
Tier 4: Honest Refusal	Out-of-Corpus Refusal Gate	2.0%	\$0.0000	0.45 ms
System Aggregate	Five-Tier Resolution Ladder	100.0%	\$0.1798	758.12 ms

5 Experimental Setup

5.1 Benchmark and Grouped Partitioning

We evaluate the framework on ClaimSafe-BN, containing 20,112 curated safety and agronomic inquiries. All items are partitioned into grouped train, development, and test splits (60/20/20 proportion) using split seed 20260813. Grouping is keyed on the semantic intent root, ensuring that all dialectal and romanized variants of an inquiry reside exclusively within the same split to prevent evaluation leakage.

5.2 Evaluation Protocols (E1–E9)

The evaluation comprises nine prespecified experiments:

- **E1 (Relational Verification):** Evaluates dangerous non-abstention on unsupported dosage claims vs. lexical baselines.
- **E2 (Calibration Tradeoff):** Measures selective risk and area under the risk-coverage curve (AURC).
- **E3 (Field Ablations):** Quantifies performance impact upon removing individual schema slots.
- **E4–E6 (Dialect Robustness):** Tests Recall@10 under standard, regional, and Banglish registers with safety non-inferiority constraints.
- **E7 (Edge Vision Parity):** Validates top-1 agreement and inference latency of exported ONNX models.

6 Results

6.1 Resolution Ladder Efficiency and Serving Economics

Table 2 summarizes empirical performance across the five resolution tiers.

Deterministic tiers (T0–T2) successfully resolve 7.8% of total traffic without invoking LLM generation, delivering sub-millisecond p95 latency (≤ 0.94 ms) and reducing effective serving costs to \$0.1798 per 1,000 queries—a 92.2% cost reduction relative to full generative cloud baselines.

6.2 On-Device Mobile Vision Parity and Latency

Table 3 reports accuracy parity and latency across on-device INT8 ONNX models and PyTorch baselines.

Table 3: **On-Device ONNX vs. PyTorch Classification Parity and Inference Latency.**

Model	Params	ONNX Size	Top-1 Parity	PyTorch Latency	ONNX Latency
crop_classifier	1.53M	5.90 MB	100.0%	38.45 ms	29.11 ms
potato_disease	5.44M	20.79 MB	100.0%	62.10 ms	47.79 ms

Table 4: **Verified Agronomic Fact Base Composition and Build-Time Dose Validations.**

Crop	Target Pathogen / Pest	Active Ingredient	Dose Range	Unit	PHI	Source Citation
Potato	Late Blight (<i>P. infestans</i>)	Mancozeb 80 WP	2.0 – 2.0	g/L	14 days	BARI Potato Hand-book 2020
Potato	Late Blight (<i>P. infestans</i>)	Metalaxyl + Mancozeb	1.5 – 2.0	g/L	14 days	BARI Plant Pathology 2022
Maize	Fall Armyworm (<i>S. frugiperda</i>)	Spinosad 45 SC	0.4 – 0.4	ml/L	14 days	DAE & BARI FAW Guide 2021
Rice	Rice Blast (<i>M. oryzae</i>)	Tricyclazole 75 WP	0.75 – 0.75	g/L	21 days	BRRI Ad-hunik Dhan 2024
Rice	Brown Planthopper (<i>N. lugens</i>)	Pymetrozine 50 WDG	0.6 – 0.6	g/L	21 days	BRRI Pest Guide 2023

Both exported ONNX models exhibit 100.0% top-1 agreement against PyTorch weights, with on-device latency well within the 150 ms mobile budget.

6.3 Multi-Crop Fact Base Generalization

Table 4 demonstrates the extension of the structured fact base across Potato, Maize, and Rice.

All fact base rows successfully passed automated cryptographic validation and dosage boundary assertions against national pesticide registries.

6.4 Safety Invariance and Golden Replay Audit

Continuous integration replay across 50 pinned golden cases achieved a 100% invariant compliance rate (0 safety breaches, 100% unanswerable query abstention). The entire backend test suite maintains a **542 passed / 0 failed** status.

7 Discussion

7.1 Relational Binding vs. Substring Matching

The critical vulnerability of lexical dosage verification lies in its inability to enforce relational binding. A lexical matcher confirms the coexistence of chemical and numeric tokens in retrieved text but cannot determine whether the generated dosage was bound to the correct crop stage or formulation. By structuring claims into typed relational tuples, KrishokChat eliminates false acceptances on misbound claims.

7.2 Linguistic Equity and Normalization Constraints

Evaluating dialectal normalization under an explicit safety-preservation constraint ensures that regional farming communities receive accessible digital advisory without suffering degraded safety guarantees.

8 Limitations

We explicitly note the bounds of this work: (1) knowledge node coverage is bounded by the 2,135 curated national research documents; (2) vision classifiers perform crop/disease classification only and do not perform bounding-box localization; and (3) this study evaluates technical verification and system safety, making no longitudinal causal claims regarding field-level farmer economic returns.

9 Ethical Considerations and Governance

KrishokChat enforces privacy-by-design through strictly local-only audit logging with automated PII regex scrubbing. Agrochemical inquiries involving acute poisoning hazards or banned substances are intercepted deterministically and routed to the national helpline (16123) and emergency healthcare contacts.

10 Reproducibility and Data Availability

All evaluation splits, benchmark datasets, and model export scripts are versioned with SHA-256 digests in the repository under `research_artifacts/` and `dataset_release/`.

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